From equation (5.7) and 5.8,

$$\vec{v}_{CM} = \frac{\sum m_i \vec{v}_i}{\sum m_i} = 0 \text{ (or) } \vec{v}_{CM} = \text{constant}$$

It implies

$$\vec{a}_{CM} = \frac{\sum m_i \vec{a}_i}{\sum m_i} = 0$$

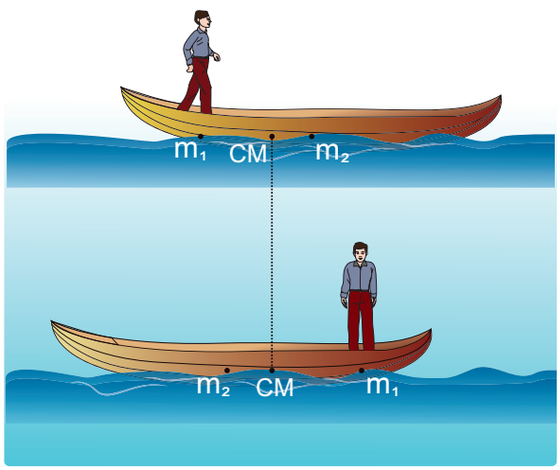
Here, the individual particles may still move with their respective velocities and accelerations due to internal forces.

In the presence of external force, (i.e. $\vec{F}_{ext} \neq 0$), the centre of mass of the system will accelerate as given by the following equation.

$$\vec{F}_{ext} = \left(\sum m_i\right)\vec{a}_{CM}; \quad \vec{F}_{ext} = M\vec{a}_{CM}; \quad \vec{a}_{CM} = \frac{\vec{F}_{ext}}{M}$$

EXAMPLE 5.5

A man of mass 50 kg is standing at one end of a boat of mass 300 kg floating on still water. He walks towards the other end of the boat with a constant velocity of 2 m s^{-1} with respect to a stationary observer on land. What will be the velocity of the boat, (a) with respect to the stationary observer on land? (b) with respect to the man walking in the boat?



[Given: There is friction between the man and the boat and no friction between the boat and water.]

Solution

Mass of the man (m_1) is, $m_1 = 50$ kg

Mass of the boat (m_2) is, $m_2 = 300$ kg

With respect to a stationary observer:

The man moves with a velocity, $v_1 = 2\ m\ s^{-1}$ and the boat moves with a velocity v_2 (which is to be found)

(i) To determine the velocity of the boat with respect to a stationary observer on land:

As there is no external force acting on the system, the man and boat move due to the friction, which is an internal force in the boat-man system. Hence, the velocity of the centre of mass is zero ($v_{CM} = 0$).

Using equation 5.7,

$$0 = \frac{\sum m_i v_i}{\sum m_i} = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2}$$

$$0 = m_1 v_1 + m_2 v_2$$

$$-m_2 v_2 = m_1 v_1$$

$$v_2 = -\frac{m_1}{m_2} v_1$$

$$v_2 = -\frac{50}{300} \times 2 = -\frac{100}{300}$$

$$v_2 = -0.33\ m\ s^{-1}$$

The negative sign in the answer implies that the boat moves in a direction opposite to that of the walking man on the boat to a stationary observer on land.

(ii) To determine the velocity of the boat with respect to the walking man:

We can find the relative velocity as,

$$v_{21} = v_2 - v_1$$

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